Tanmay Goel

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Data & AI Engineer — building RAG agents, multi-agent orchestration and production data pipelines.

Skills

Languages : Python, TypeScript, SQL, JavaScript, Rust*
LLM & Agents : Vercel AI SDK, smolagents, Anthropic managed agents, LangChain TextSplitters, tool-calling, RAG
Data & Storage : PostgreSQL (Neon, pgvector, HNSW), TimescaleDB, MongoDB, Cloudflare D1/R2, Redis (Upstash), QStash
ML & Analytics : pandas, NumPy, SciPy, yfinance, PyPortfolioOpt, matplotlib
Backend & Infra : Node.js, Next.js, Hono, FastAPI*, Flask*, Docker, REST, WebSocket
Cloud & Tooling : GCP, AWS*, Cloudflare Workers, Vercel, Git, GitHub Actions, Vitest, Linux, Bash

Education

Harcourt Butler Technical University (HBTU) — *Kanpur, India*
Integrated Master's in Mathematics & Data Science — *Expected May 2030*

Experience

Data Engineer | *Taleoid Page — Chennai, India — Remote* — Jan 2026 – Mar 2026

- Replaced manual Excel tracking with **Python and Oracle SQL** pipelines for **380+ active users** (700 total accounts)
- Built **demand and popularity scoring** over purchase history to rank new titles and surface borrowing demand
- Engineered automated **email notification** pipelines for return reminders, demand alerts and eBook licence expiry

Projects

Destr | *Next.js 16, Vercel AI SDK, Drizzle, pgvector, Clerk* — github.com/tanmay442/Destr

- Built a **serverless** tool-calling RAG knowledge agent on Next.js 16 + Vercel AI SDK v6, running on **Neon Postgres**, Clerk and Upstash
- Owned the **chunking and retrieval layer**: **5 pluggable chunking strategies** (semantic, parent-child, and more) with configurable retrieval — vector vs. agentic mode, hybrid search, cross-encoder rerank
- Implemented **agentic retrieval**: an empty result triggers a rewrite-and-retry pass before an out-of-domain wall, with **grounded PDF citations**, hallucination checks and automatic ticket escalation
- Built a **golden-dataset eval harness** with a 0.7 faithfulness gate, **weekly CI** runs, a **daily cron**, and a **Redis** cache

FinSmolAgent | *FastAPI, smolagents, TimescaleDB, WebSocket, React* — github.com/tanmay442/FinSmolAgent

- Architected a **six-stage multi-agent pipeline** — KYC/AML screening, market research and earnings review in parallel, DCF modelling, sandboxed backtest, valuation reporting — across **5 agents** and 22 composable skills
- Enforced a **three-tier trust boundary** with **JSON-Schema-validated handoffs**, provenance on every metric, and a **compliance approval gate** that blocks execution until a human decision
- Ran backtests inside a **sandboxed worker** with a workspace jail, hard timeout kill, deterministic seeds and a package whitelist, streamed live over **FastAPI WebSocket** events to a React console

HackNova Equity Analytics | *Python, pandas, SciPy, yfinance, Streamlit* — github.com/tanmay442/hacknova

- Built an end-to-end **equity analytics pipeline** over **15 NSE stocks** plus the Nifty 50 benchmark (2023-2024): Yahoo Finance ingestion, trading-date alignment, and gap imputation dropping assets with 5+ day gaps
- Computed risk and return analytics — annualised return, volatility, **Sharpe ratio** at a 6.5% risk-free rate and beta versus the index — with a correlation heatmap and per-ticker SMA crossover signal tables
- Added a SciPy-optimised portfolio layer comparing an equal-weight baseline against a Golden-Cross-filtered max-Sharpe portfolio, a **-10% market-crash** stress test, and a Streamlit dashboard

Telegram Agent | *Python, OpenRouter, SQLite, OAuth 2.0* — github.com/tanmay442/Telegram_Agent

- Built a bot service on python-telegram-bot handling image and PDF compression and conversion (Pillow, PyMuPDF, img2pdf), driven by a per-user **state machine** with request rate limits
- Added **OpenRouter** LLM chat that extracts PDF context from uploads (pypdf) and normalises history into chat-completions payloads with retry and backoff, plus **Google Workspace** over OAuth 2.0

Achievements

- **1st place, Smart India Hackathon (SIH) 2026 — HBTU internal college hackathon round**
- **Philosophy Spread — Logic Module 04**: 160k+ unique page visits